PART I - FOUNDATION PLC PROJECTS

Project 7 - Automated Batch Mixing System

Version 1 - Foundation PLC Project

PLC Platform: Allen-Bradley MicroLogix 1100

Programming Software: RSLogix Micro Starter Lite 8.30

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1. Project Overview

The Automated Batch Mixing System was developed to introduce step-based sequential control using internal PLC memory bits, timers, valves, and a mixer motor.

The visible Version 1 sequence begins when B3:0/0 Cycle_Active is true. The program then progresses automatically through four operating steps: Fill A, Fill B, Mix, and Drain. Each step is represented by a dedicated internal bit, controls its associated output, and uses a timer to determine when the sequence should advance.

Fill A and Fill B each operate for five seconds, the mixer runs for ten seconds, and the drain step operates for five seconds. At the end of the drain period, the Drain step and Cycle_Active bit are unlatched, returning the visible sequence to its idle condition.

This project represents a significant progression from earlier single-function programs because the PLC coordinates several outputs through a defined sequence of internal states and timer-based transitions.

2. Engineering Objective

The objective of this project was to develop and verify a PLC-controlled batch sequence that automatically progressed through ingredient filling, mixing, and draining using internal step bits and timer-based transitions.

- Initialize the Fill A step when the cycle is active and no other step is selected.
- Open Ingredient A Valve during the Fill A step for five seconds.
- Transition automatically from Fill A to Fill B.
- Open Ingredient B Valve during the Fill B step for five seconds.
- Transition automatically from Fill B to Mix.
- Run the Mixer Motor during the Mix step for ten seconds.
- Transition automatically from Mix to Drain.
- Open the Drain Valve during the Drain step for five seconds.
- Clear the Drain step and Cycle_Active bit at the end of the batch.

3. Functional Requirements

- B3:0/0 shall represent Cycle_Active for the visible batch sequence.
- When Cycle_Active is true and no step bit is active, B3:0/1 Fill_A shall latch ON.
- Ingredient A Valve O:2/0 shall energize while Cycle_Active and Fill_A are true.
- Timer T4:0 shall time the Fill A step for five seconds.
- When T4:0/DN becomes true, Fill_A shall unlatch and B3:0/2 Fill_B shall latch.
- Ingredient B Valve O:2/1 shall energize while Cycle_Active and Fill_B are true.
- Timer T4:1 shall time the Fill B step for five seconds.
- When T4:1/DN becomes true, Fill_B shall unlatch and B3:0/3 Mix shall latch.
- Mixer Motor O:2/2 shall energize while Cycle_Active and Mix are true.
- Timer T4:2 shall time the Mix step for ten seconds.
- When T4:2/DN becomes true, Mix shall unlatch and B3:0/4 Drain shall latch.
- Drain Valve O:2/3 shall energize while Cycle_Active and Drain are true.
- Timer T4:3 shall time the Drain step for five seconds.
- When T4:3/DN becomes true, Drain and Cycle_Active shall both unlatch.

- The report shall not invent the rung that originally establishes Cycle_Active because that rung is not shown in the supplied screenshots.

4. System Components

Component	Identification	Function
PLC Platform	Allen-Bradley MicroLogix 1100	Executes the batch sequence
Cycle Active	B3:0/0	Enables the visible batch cycle
Fill A Step	B3:0/1	Represents Ingredient A filling
Fill B Step	B3:0/2	Represents Ingredient B filling
Mix Step	B3:0/3	Represents mixing operation
Drain Step	B3:0/4	Represents draining operation
Ingredient A Valve	O:2/0	Controls Ingredient A flow
Ingredient B Valve	O:2/1	Controls Ingredient B flow
Mixer Motor	O:2/2	Performs batch mixing
Drain Valve	O:2/3	Drains the completed batch
Fill A Timer	T4:0	Times Ingredient A fill
Fill B Timer	T4:1	Times Ingredient B fill
Mix Timer	T4:2	Times mixing operation
Drain-Step Timer	T4:3	Times draining operation
Programming Software	RSLogix Micro Starter Lite 8.30	Ladder logic development and monitoring

5. I/O and Data Assignment

Physical Outputs

Device	PLC Address	Type	Description
Ingredient A Valve	O:2/0	Digital Output	Ingredient A fill command
Ingredient B Valve	O:2/1	Digital Output	Ingredient B fill command
Mixer Motor	O:2/2	Digital Output	Mixer command
Drain Valve	O:2/3	Digital Output	Drain command

Internal Sequence Bits

Function	Address	Purpose
Cycle Active	B3:0/0	Enables the batch sequence
Fill A	B3:0/1	Active Fill A step
Fill B	B3:0/2	Active Fill B step
Mix	B3:0/3	Active mixing step
Drain	B3:0/4	Active draining step

Timer Assignment

Timer	Time Base	Preset	Resulting Time	Sequence Function
T4:0	0.01 s	500	5.00 s	Fill A
T4:1	0.01 s	500	5.00 s	Fill B
T4:2	0.01 s	1000	10.00 s	Mix
T4:3	0.01 s	500	5.00 s	Drain

In the supplied screenshot, T4:3 is displayed with the symbolic text 'Mixer Timer' even though it is enabled by the Drain step and is used to end the drain sequence. This report preserves the actual address T4:3 and describes its functional role in the visible logic as the drain-step timer.

6. Control Strategy

I used a step-based control strategy in which one internal bit represents the active stage of the batch. Each step controls its own output and timer, and each timer Done bit performs the transition to the next step.

Rung 0000 initializes Fill_A only when Cycle_Active is true and Fill_A, Fill_B, Mix, and Drain are all false. This prevents the initializer from selecting Fill_A while another visible step is already active.

At the end of each timed step, the current step bit is unlatched and the next step bit is latched. This creates a clear progression from Fill A to Fill B to Mix to Drain.

The last transition differs from the intermediate transitions: when T4:3 completes, both Drain and Cycle_Active are unlatched. This ends the visible batch cycle rather than starting another process step.

7. Engineering Design Decisions

Design Decision	Reason
Used dedicated B3 step bits	Made each batch stage explicit and easy to monitor
Used separate output and timer rungs for each step	Improved readability and troubleshooting
Used timer Done bits for transitions	Provided automatic, deterministic progression between timed stages
Unlatched current step before latching next step	Maintained a clear one-step-at-a-time sequence
Used XIO checks on all step bits in the initializer	Allowed Fill A to initialize only when no visible step was active
Cleared Cycle_Active after Drain	Returned the visible sequence to idle after one completed batch
Kept Version 1 timer-based	Provided a clear foundation sequence without adding sensor confirmations or fault diagnostics

The current screenshots do not show how Cycle_Active B3:0/0 is initially set. I therefore treat Cycle_Active as the enabling condition for the visible sequence and do not invent a Start rung that is not present in the supplied implementation.

8. Ladder Logic Description

The supplied screenshots show thirteen functional rungs, Rung 0000 through Rung 0012. The representations below follow the approved single-line portfolio format.

8.1 Rung 0000 - Initialize Fill A Step

```
|--[ XIC B3:0/0 ]--[ XIO B3:0/1 ]--[ XIO B3:0/2 ]--[ XIO B3:0/3 ]--[ XIO B3:0/4 ]--( OTL B3:0/1 )--|
    Cycle_Active       Fill_A          Fill_B           Mix            Drain            Fill_A
```

When Cycle_Active is true and none of the four step bits is active, B3:0/1 Fill_A is latched.

8.2 Rung 0001 - Ingredient A Valve

```
|----[ XIC B3:0/0 ]----[ XIC B3:0/1 ]------------------------( OTE O:2/0 )----|
      Cycle_Active        Fill_A                            Ingredient_A_Valve
```

Ingredient A Valve O:2/0 energizes during the Fill A step.

8.3 Rung 0002 - Fill A Timer

```
|----[ XIC B3:0/0 ]----[ XIC B3:0/1 ]------------------------[ TON T4:0 ]----|
     Cycle_Active          Fill_A                            Fill_A_Timer  PRE: 500  Time Base: 0.01
```

T4:0 times the Fill A step for five seconds.

8.4 Rung 0003 - Transition Fill A to Fill B

```
|----[ XIC B3:0/0 ]----[ XIC T4:0/DN ]------------------------+----( OTU B3:0/1 )----|
      Cycle_Active      Fill_A_Timer_Done                      |      Fill_A
                                                               |
                                                               +----( OTL B3:0/2 )----|
                                                                      Fill_B
```

When T4:0/DN becomes true, Fill_A is unlatched and Fill_B is latched.

8.5 Rung 0004 - Ingredient B Valve

```
|----[ XIC B3:0/0 ]----[ XIC B3:0/2 ]------------------------( OTE O:2/1 )----|
      Cycle_Active         Fill_B                            Ingredient_B_Valve
```

Ingredient B Valve O:2/1 energizes during the Fill B step.

8.6 Rung 0005 - Fill B Timer

```
|----[ XIC B3:0/0 ]----[ XIC B3:0/2 ]------------------------[ TON T4:1 ]----|
       Cycle_Active         Fill_B                           Fill_B_Timer   PRE: 500  Time Base: 0.01
```

T4:1 times the Fill B step for five seconds.

8.7 Rung 0006 - Transition Fill B to Mix

```
|----[ XIC B3:0/0 ]----[ XIC T4:1/DN ]------------------------+----( OTU B3:0/2 )----|
      Cycle_Active      Fill_B_Timer_Done                      |      Fill_B
                                                               |
                                                               +----( OTL B3:0/3 )----|
                                                                      Mix
```

When T4:1/DN becomes true, Fill_B is unlatched and Mix is latched.

8.8 Rung 0007 - Mixer Motor

```
|----[ XIC B3:0/0 ]----[ XIC B3:0/3 ]------------------------( OTE O:2/2 )----|
       Cycle_Active         Mix                               Mixer_Motor
```

Mixer Motor O:2/2 energizes during the Mix step.

8.9 Rung 0008 - Mixing Timer

```
|----[ XIC B3:0/0 ]----[ XIC B3:0/3 ]------------------------[ TON T4:2 ]----|
      Cycle_Active         Mix                              Mixing_Timer   PRE: 1000  Time Base: 0.01
```

T4:2 times the Mix step for ten seconds.

8.10 Rung 0009 - Transition Mix to Drain

```
|----[ XIC B3:0/0 ]----[ XIC T4:2/DN ]------------------------+----( OTU B3:0/3 )----|
      Cycle_Active      Mixing_Timer_Done                      |      Mix
                                                               |
                                                               +----( OTL B3:0/4 )----|
                                                                      Drain
```

When T4:2/DN becomes true, Mix is unlatched and Drain is latched.

8.11 Rung 0010 - Drain Valve

```
|----[ XIC B3:0/0 ]----[ XIC B3:0/4 ]------------------------( OTE O:2/3 )----|
      Cycle_Active        Drain                                Drain_Valve
```

Drain Valve O:2/3 energizes during the Drain step.

8.12 Rung 0011 - Drain-Step Timer

```
|----[ XIC B3:0/0 ]----[ XIC B3:0/4 ]------------------------[ TON T4:3 ]----|
      Cycle_Active        Drain                                 T4:3   PRE: 500   Time Base: 0.01
```

T4:3 times the drain step for five seconds. The screenshot displays the symbolic label 'Mixer Timer' for T4:3; functionally, this timer is enabled by Drain and is used by the end-cycle rung.

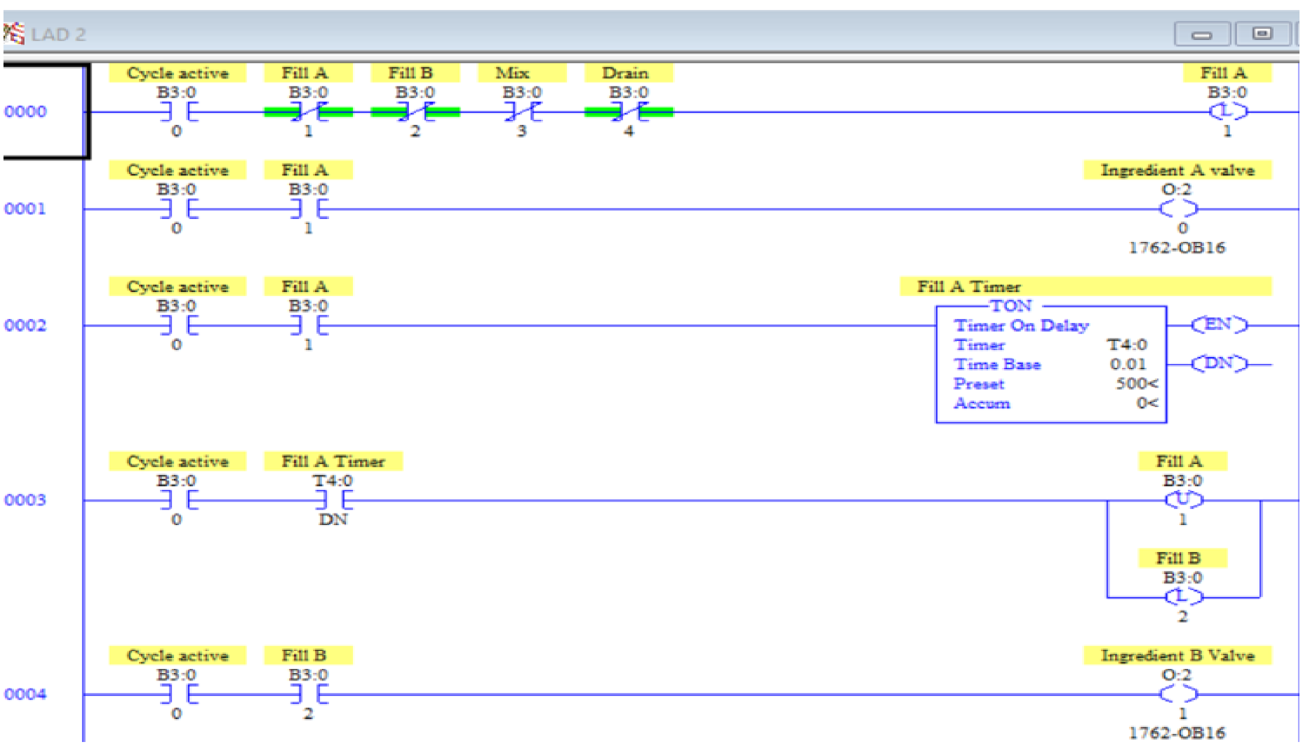
8.13 Rung 0012 - End Batch Cycle

```
|----[ XIC B3:0/0 ]----[ XIC T4:3/DN ]-----------------------+----( OTU B3:0/4 )----|
       Cycle_Active        T4:3_Done                         |      Drain
                                                             |
                                                             +----( OTU B3:0/0 )----|
                                                                    Cycle_Active
```

When T4:3/DN becomes true, Drain and Cycle_Active are both unlatched, ending the visible batch cycle.

8.14 Actual RSLogix Implementation

The three figures below are the supplied RSLogix screenshots used as the technical source of truth for this report.



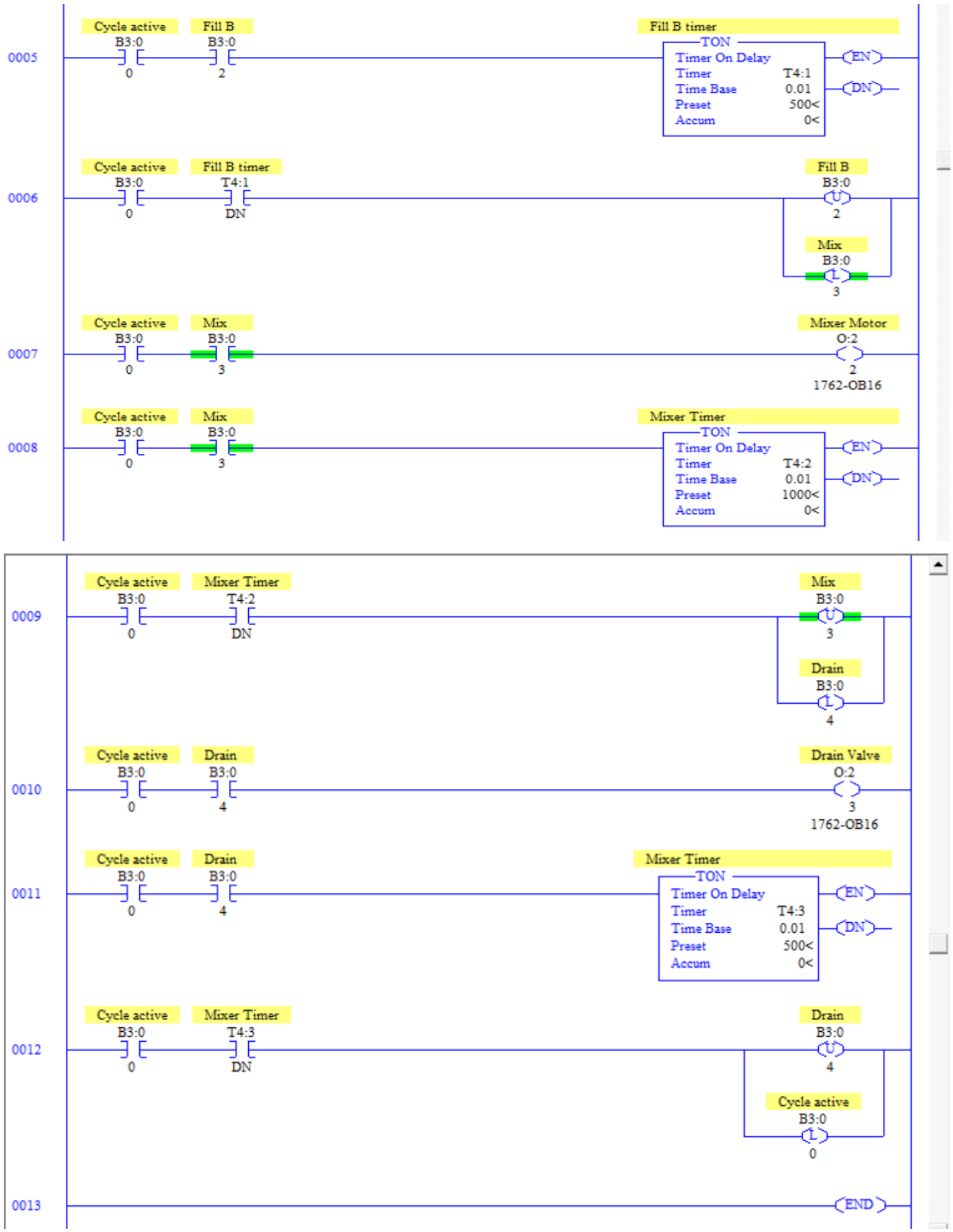


Figure P713 - RSLogix Batch Mixing Logic

9. Sequence of Operation

Step 1 - Cycle Active: B3:0/0 is true, enabling the visible batch sequence.

Step 2 - Fill A Initialized: If no other step is active, B3:0/1 Fill_A is latched.

Step 3 - Ingredient A Fill: O:2/0 opens and T4:0 times for five seconds.

Step 4 - Transition to Fill B: T4:0/DN unlatches Fill_A and latches Fill_B.

Step 5 - Ingredient B Fill: O:2/1 opens and T4:1 times for five seconds.

Step 6 - Transition to Mix: T4:1/DN unlatches Fill_B and latches Mix.

Step 7 - Mixing: O:2/2 runs and T4:2 times for ten seconds.

Step 8 - Transition to Drain: T4:2/DN unlatches Mix and latches Drain.

Step 9 - Draining: O:2/3 opens and T4:3 times for five seconds.

Step 10 - End Cycle: T4:3/DN unlatches Drain and Cycle_Active, returning the visible sequence to idle.

10. Testing & Validation

The supplied ladder screenshots represent the final tested Version 1 implementation. The following tests verify the visible sequence and are recorded as PASS.

Test ID	Test Action / Condition	Expected Result	Observed Result	Status
T-01	Cycle_Active true with no step active	Fill_A latches	Fill_A latched	PASS
T-02	Fill_A active	Ingredient A Valve ON; T4:0 timing	Valve energized; timer operated	PASS
T-03	T4:0 reaches 5 s	Fill_A clears; Fill_B latches	Transition completed	PASS
T-04	Fill_B active	Ingredient B Valve ON; T4:1 timing	Valve energized; timer operated	PASS
T-05	T4:1 reaches 5 s	Fill_B clears; Mix latches	Transition completed	PASS
T-06	Mix active	Mixer Motor ON; T4:2 timing	Mixer energized; timer operated	PASS
T-07	T4:2 reaches 10 s	Mix clears; Drain latches	Transition completed	PASS
T-08	Drain active	Drain Valve ON; T4:3 timing	Drain valve energized; timer operated	PASS
T-09	T4:3 reaches 5 s	Drain clears; Cycle_Active clears	Batch cycle ended	PASS
T-10	Complete visible sequence	Steps occur Fill A -> Fill B -> Mix -> Drain -> Idle	Sequence completed in order	PASS

Validation Summary

Testing confirmed that each step controlled the correct output and timer, that each timer Done bit transferred control to the next step, and that the end-cycle rung cleared both Drain and Cycle_Active.

All ten defined Version 1 functional tests passed. The validation is limited to the supplied PLC project implementation and does not represent industrial commissioning or safety validation.

11. Results

The completed Automated Batch Mixing System successfully progressed through four timed operating stages: Fill A, Fill B, Mix, and Drain.

Each stage used a dedicated internal step bit, output command, and timer. Timer Done bits correctly transferred the active state from one step to the next.

At the end of the drain period, the program cleared both the Drain step and Cycle_Active, ending the visible batch cycle. All ten defined functional tests passed.

12. Challenges & Troubleshooting

A useful troubleshooting path for this project is: Cycle_Active -> Active Step Bit -> Output -> Step Timer -> Timer Done -> Next Step Transition.

Symptom	Possible Cause	Diagnostic Check	Corrective Action
Fill A does not initialize	Cycle_Active false or another step bit remains active	Monitor B3:0/0 through B3:0/4	Verify initializer conditions
Ingredient A Valve does not open	Fill_A not active or Cycle_Active false	Monitor B3:0/1 and O:2/0	Verify Fill A step and output rung
Sequence does not move to Fill B	T4:0/DN not true	Monitor T4:0.ACC and T4:0/DN	Verify Fill A timer and transition rung
Ingredient B Valve does not open	Fill_B not active	Monitor B3:0/2 and O:2/1	Verify Fill B transition and output rung
Mixer does not start	Mix bit not active	Monitor B3:0/3 and O:2/2	Verify T4:1/DN transition
Drain does not start	Drain bit not active	Monitor B3:0/4 and O:2/3	Verify T4:2/DN transition
Cycle does not end	T4:3/DN not true or end-cycle rung not executing	Monitor T4:3 and Rung 0012	Verify drain-step timer and unlatch logic
More than one step appears active	Previous step was not unlatched during transition	Monitor B3:0/1 through B3:0/4	Verify OTU/OTL transition rung

The project reinforced the importance of monitoring step bits during sequence troubleshooting. If a process output does not behave correctly, the first question is whether the expected step is actually active.

13. Lessons Learned

This project helped me understand how internal memory bits can be used as explicit sequence states rather than only as general-purpose control flags.

I learned how timer Done bits can perform deterministic transitions from one stage of a batch to the next.

Separating each output, timer, and transition into dedicated rungs made the sequence easier to follow than combining several operations into one rung.

I also learned that end-of-cycle logic is part of the sequence design. Clearing the final step and Cycle_Active returns the visible batch sequence to a known idle condition.

14. Industrial Relevance

Step-based batch sequences are common in mixing, blending, food processing, chemical preparation, dosing, and other process-automation applications.

The Version 1 project demonstrates the foundation of a batch process in which ingredients are added in order, mixed for a defined duration, and then discharged.

A real industrial batch system would normally include level confirmation, valve feedback, motor feedback, material availability checks, permissives, recipe values, fault handling, manual controls, operator displays, batch records, and safety-related interlocks where required.

This Version 1 project should therefore be viewed as a foundation sequence-control application rather than a production-ready batch system.

15. Transition to Industrial Version (Version 2)

Version 2 can retain the same basic Fill A -> Fill B -> Mix -> Drain sequence while making the machine state, permissives, diagnostics, and recovery behavior more explicit.

Version 2 Improvement	Engineering Benefit
Explicit Start/Stop Run Request	Defines how Cycle_Active is initiated and stopped
Auto/Manual operating modes	Supports automatic batch operation and maintenance control
Mode-conflict detection	Identifies invalid simultaneous mode commands
Step permissives	Prevents a step from starting unless required conditions are satisfied
Valve feedback	Verifies commanded ingredient/drain valve position
Mixer feedback	Verifies motor response
Level / quantity confirmation	Confirms actual filling rather than relying only on time
Material availability checks	Prevents empty-source filling attempts
Step timeout faults	Detects stages that do not complete as expected
Sequence fault latch and reset	Provides controlled recovery after abnormal operation
Recipe / adjustable timing values	Supports different batch formulations
HMI sequence display	Shows current step, timers, outputs, alarms, and status
Batch complete indication	Provides explicit operator feedback
Batch records / SCADA integration	Supports production history and reporting

A stronger Version 2 architecture could become: Start Request -> Cycle Active -> Permissive Check -> Fill A -> Verify -> Fill B -> Verify -> Mix -> Verify -> Drain -> Verify -> Batch Complete -> Reset/Ready.

The Version 2 design should preserve the clarity of the current step-based sequence while adding feedback, diagnostics, and controlled abnormal-condition handling.

16. Project Summary

The Automated Batch Mixing System introduced a structured step-based PLC sequence using internal bits, timers, valves, and a mixer motor.

With Cycle_Active B3:0/0 true, the program initialized Fill_A and progressed automatically through Fill A for five seconds, Fill B for five seconds, Mix for ten seconds, and Drain for five seconds.

Each timer Done bit cleared the current step and selected the next step. At the end of the Drain step, T4:3/DN cleared both Drain and Cycle_Active, ending the visible batch cycle.

All ten defined functional tests passed. The project strengthened my understanding of sequence states, timer-driven transitions, step-based output control, and end-of-cycle logic, and it provides a strong foundation for a more complete industrial batch-processing design.